

Viral Hepatitis

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Introduction

- Hepatitis is the inflammation and damage of hepatocytes in the liver caused by
 - Hepatitis A, B, C, D, E viruses
 - Alcohol, and
 - Medications
- Viral hepatitis occurs at any age and is the most common cause of liver disease.
- The prevalence and incidence may be underreported:
 - because patients are often asymptomatic.

Introduction.....

- ❖ Acute hepatitis may occur with all five types of viral hepatitis and..... < 6 months in duration.
- ❖ Chronic viral hepatitis (>6 months duration) may lead to→→ cirrhosis→→end-stage liver disease (ESLD) →→hepatocellular carcinoma(HCC).
- ❖ Chronic viral hepatitis is associated with hepatitis B, C, and D and may lead to the development of
 - Ascites
 - Jaundice
 - Hepatic encephalopathy
 - Esophageal varices
 - Cirrhosis, and
 - Hepatocellular carcinoma.

Hepatitis A

- Hepatitis A virus (HAV) is an acute viral infection spread via the fecal-oral route.
- In the US, HAV has declined considerably...b/c of HAV vaccine development.
 - but HAV remains prevalent in underdeveloped countries.
- HAV impacts areas with inadequate sanitation and poor personal hygienic practices.
- Other risk factors include:
 - IV drug use
 - Exposure to infected individuals
 - Homosexual activity in men
 - International travelers to endemic areas
 - Patients with clotting factor disorders

Hepatitis B

- Hepatitis B virus (HBV) causes an acute and chronic viral infection.
- Acute HBV is transmitted through
 - ✓ Exposure to blood and bodily secretions
 - ✓ Sexual activity
 - ✓ IV drug use
 - ✓ Perinatal transmission or Infants born from infected mothers
 - ✓ Occupational exposure.
- ~2 billion people worldwide have evidence of past or present HBV infection.
- Globally, more than 680,000 deaths are associated with HBV annually.
- ~10% to 15% of patients with HBV infection develop chronic HBV disease.
 - ❖ Therefore, 85% to 90% of acute HBV infections resolve without complications.

Hepatitis C

- Hepatitis C virus (HCV) infection is the most common blood-borne infection.
- 80 million people worldwide have chronic HCV
- ~10% to 15% of patients with acute HCV resolve without any further sequelae;
 - ❖ Therefore, 85% to 90% of HCV cases develop into chronic disease.
- There are six geographically specific genotypes of HCV
 - Genotype 1 represents 75% of infections within the US
 - Genotype 4 is common in the Middle East, Egypt and Sub-Saharan Africa
 - HCV genotype is used to determine the duration of therapy and therapeutic response

Hepatitis C

- The most common modes of transmission: IV drug use and blood transfusion (before July 1992)
- Other HCV risk factors:
 - Tattoos and Body piercings
 - Occupational exposures
 - HIV infection
 - Chronic hemodialysis
 - Person born from 1945 through 1965 (Baby boomers)
 - Multiple sexual partners though rare through in heterosexual monogamous r/ships.

Hepatitis D

- Hepatitis D affects up to 20 million people worldwide
- HDV infection is unique b/c it can only occur in people who are also infected with the HBV.
- The most likely modes of transmitting HDV are similar to those of HBV

Hepatitis E

- An estimated 20 million individuals worldwide are infected annually with HEV.
- The disease was associated with about 44,000 deaths in 2015.
- HEV infection primarily occurs in underdeveloped countries with poor sanitation.
- The HEV is transmitted mainly by the fecal–oral route
eg, contaminated drinking water

Pathophysiology

- Hepatitis A.... a non-enveloped single-stranded RNA virus
- HAV infection is usually acute, self-limiting, and Immunity after attack probably lasts for life.
- HAV's life cycle in the human host
- Ingestion of the virus →→Absorption in the stomach or small intestine→→entry into the circulation →→uptake by the liver→→Replication within hepatocytes and GI epithelial cells.
- New virus particles are released into the blood and secreted into bile by the liver.
- The virus is then either reabsorbed to continue its cycle or excreted in the stool.
- The enterohepatic cycle will continue until interrupted by antibody neutralization

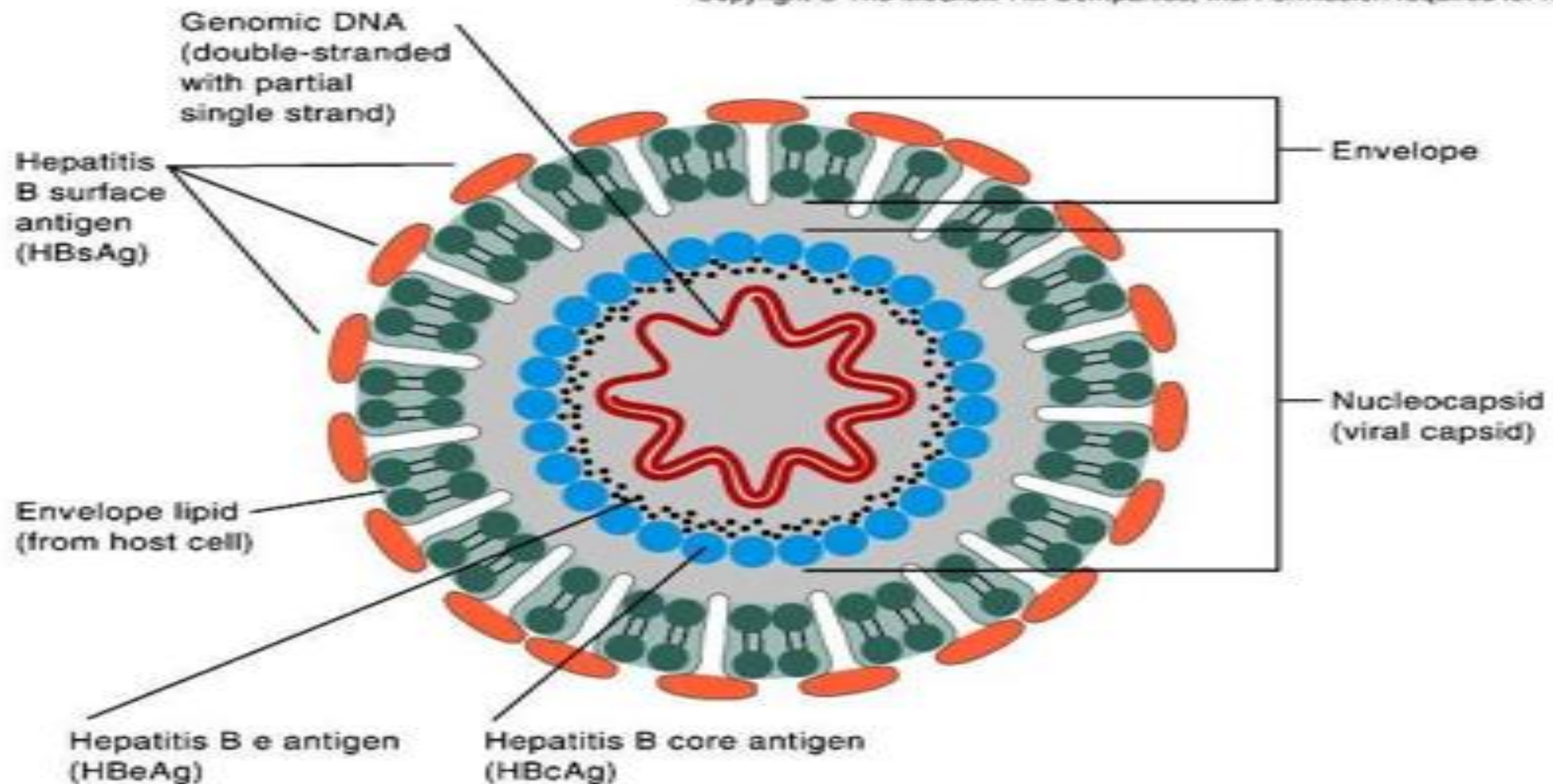
Pathophysiology

Hepatitis B

- DNA virus with a phospholipid layer containing hepatitis B surface antigen (HBsAg) that surrounds the nucleocapsid.
- The nucleocapsid contains the core protein that produces hepatitis B core antigen (HBcAg).
- Hepatic injury is due to a cytotoxic immune reaction when HBcAg is expressed on hepatic cells surface.
- Fortunately, antibodies against hepatitis B core antigen (anti-HBc) are measurable in the blood.
 - where anti-HBc to immunoglobulin M (IgM) indicates active infection, and
 - anti-HBc to IgG indicates either chronic infection or possible immunity against HBV.
- Viral replication occurs if hepatitis B envelope antigen (HBeAg) is present and circulate in the blood.
- The serum HBV DNA concentration is a measure of viral infectivity and quantifies viral replication.

❖ HBV : Structure

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Pathophysiology.....

- Once the HBV infection resolves,
 - antibodies against the hepatitis B envelope (anti-HBe)
 - antibodies against hepatitis B surface antigen (anti-HBs) develop and
 - HBV DNA levels become undetectable.
- If the antibodies do not develop, the likelihood of developing Chronic Hepatitis B (CHB) increases.
- This depends primarily on the host's immune system at the time of infection.
 - In immunocompetent individuals, the disease resolves spontaneously...doesn't lead to complications.
 - In immunocompromised persons...cause persistent infection...hepatic cell damage and inflammation.
- ~90% of adults infected with the HBV develop anti-HBs..... ..lifelong immunity.
- Unfortunately, about 90% of infants born to mothers with active HBV infection will develop CHB.

Pathophysiology

- Hepatitis C....single strand RNA virus
- In most cases, an acute HCV infection leads to chronic infection.
- HCV poses a daunting challenge for immune control because of its rapid viral diversification.
- HCV genomic mutations are detectable within 1 year of infection.
- Resolved cases of HCV are defined by a vigorous T-cell response with highly active CD8 and persistent CD4-cell response.
- CD8 activity mediates protective immunity but requires the aid of CD4 cells to maintain the response during viral mutations.
- Antibodies against HCV (anti-HCV) in the blood indicate infection.

Pathophysiology

- Hepatitis D
- RNA virus that requires the presence of HBV for replication,
 - causing either coinfection (both hepatitis B and D infection occurring simultaneously) or
 - Superinfection (acquiring HDV after having long-standing HBV disease)
- This occurs because the HDV antigen (HDVAg) is coated by the HBsAg.

Pathophysiology

Hepatitis E:

- Non-enveloped single-stranded messenger RNA virus with at least 4 genotypes;
 - Genotypes 1 and 2 are found in humans
 - Genotypes 3 and 4 occur in animals.
- The HEV like HAV virus infect people via the fecal–oral route.
- HEV infections are usually self-limiting...rarely result in hepatic complications.
- More common in immunocompromised individuals (eg, HIV infection) and organ transplant recipients.

Clinical Presentations

□ Incubation periods for viral hepatitis

- HAV: is approximately 28 days, with a range of 15 to 50 days
- HBV: 45-180 days (usually 60-90 days)
- HVC: 40-120 days
- HVD: 2-12 weeks
- HVE: 2-9 weeks

Clinical Presentation

- ❖ Signs and symptoms vary among patients
 - Range from asymptomatic to liver failure
- Asymptomatic patients may only present with mildly elevated liver enzymes.
- Common acute symptoms include:
 - ✓ Fatigue,
 - ✓ Jaundice (icteric sclera)
 - ✓ Nausea and vomiting
 - ✓ Weight loss
 - ✓ Fever
 - ✓ Right upper quadrant pain and
 - ✓ Splenomegaly.

Clinical Presentation



Clinical Presentation of Acute Hepatitis

	Hepatitis A (HAV)	Hepatitis B (HBV)	Hepatitis C (HCV)
Signs and symptoms	>70% of patients are symptomatic with fever, jaundice, scleral icterus, hepatomegaly. Less common: splenomegaly, skin rash, arthralgia.	Approximately 70% of patients are anicteric or subclinical. Younger patients most likely to be asymptomatic. If symptoms occur, jaundice, dark urine, white stool, abdominal pain, fatigue, fever, chills, loss of appetite, and pruritus are possible.	Approximately 70% of patients are asymptomatic. If symptoms occur, jaundice, dark urine, white stool, abdominal pain, fatigue, fever, chills, loss of appetite, and pruritus are possible.
<i>Laboratory Findings</i>			
Aminotransferase (ALT, AST) Elevations	>1,000 IU/L (16.7 μ kat/L) ALT>AST	1,000-2,000 IU/L (16.7 - 33.3 μ kat/L) ALT>AST	Highly variable, can be approximately 1,000 IU/L (16.7 μ kat/L) ALT>AST
Bilirubin	Elevated and preceded by aminotransferase elevations	Can be within normal or elevated	Elevated and preceded by aminotransferase elevations
Virus Specific Tests	IgM anti-HAV	IgM anti-HBc (+), HBsAg (+)	HCV RNA (+) or quantifiable; HCV antibody reactive within 12 weeks of exposure

IU, international units; L, liter; ALT, alanine transaminase; AST, aspartate transaminase; anti-, antibody; IgM, immunoglobulin; HBc, hepatitis B core; HBsAg, hepatitis B surface antigen; RNA, ribonucleic acid

Diagnosis

- Diagnosing viral hepatitis is difficult because patients may be asymptomatic.
- Symptoms cannot identify the type of hepatitis;
 - Laboratory serology must be obtained.
- Viral hepatitis is diagnosed by the presence of
 - Antibodies, antigens, and measured virus.
- Elevated liver enzymes are not specific to viral hepatitis but of hepatocellular injury.
- Liver biopsy....for assessing the grade and stage of liver disease in chronic HBV or HCV.
 - The grade reflects the degree of liver inflammation, and the stage reflects extent of fibrosis or presence of cirrhosis.

Interpretation of Viral Hepatitis Serology Panels

Type	Laboratory Test	Result	Interpretation of Panel
Hepatitis A	IgM anti-HAV	Negative	Susceptible to infection
	IgG anti-HAV	Negative	
	IgM anti-HAV	Positive	Acutely infected
	IgG anti-HAV	Positive	Immune due to either natural infection or HAV vaccine
Hepatitis B ^a	HBsAg	Negative	Susceptible to infection
	anti-HBc	Negative	
	anti-HBs	Negative	
	HBsAg	Negative	Immune due to natural infection
	anti-HBc	Positive	
	anti-HBs	Positive	
	HBsAg	Negative	Immune due to hepatitis B vaccination
	anti-HBc	Negative	
	anti-HBs	Positive	
	HBsAg	Positive	Acutely infected
	anti-HBc	Positive	
	IgM anti-HBc	Positive	
	anti-HBs	Negative	
	HBsAg	Positive	Chronically infected
	anti-HBc	Positive	
	IgM anti-HBc	Negative	
	anti-HBs	Negative	
	HBsAg	Negative	Four interpretations possible: (a) Resolved infection (most common); (b) false-positive anti-HBc, thus susceptible; (c) low-level chronic infection; (d) resolving acute infection
	anti-HBc	Positive	
	Anti-HBs	Negative	

	Anti-HBc	Negative	Low level chronic infection, (or) resolving acute infection
Hepatitis C	anti-HCV	Negative	Susceptible to infection
	anti-HCV	Positive	Acutely or chronically infected
Hepatitis D ^b	IgM anti-HDV	Positive	Acute HBV-HDV coinfection
	HDVAg	Positive	
	HBsAg	Positive	
	HBeAg	Positive	
	anti-HBc	Positive	
Hepatitis E	IgM anti-HEV	Negative	Susceptible to infection
	IgG anti-HEV	Negative	
	IgM anti-HEV	Positive	Acutely infected
	IgG anti-HEV	Positive	Immune due to natural infection

^aCenters for Disease Control and Prevention. Hepatology [Internet]. <http://www.cdc.gov/hepatitis/HBV/PDFs/SerologicChartv8.pdf>.

^bHepatitis D should be suspected in those who have HBsAg positivity. Hepatitis D may present as either coinfection where both HDV and HBV serologies appear simultaneously, whereas for superinfection, HBV has been present for some time and later HDV develops.

anti-HAV, hepatitis A antibody; anti-HBc, hepatitis B core antibody; anti-HBs, hepatitis B surface antibody; HBsAg, hepatitis B surface antigen; anti-HCV, hepatitis C antibody; anti-HDV, hepatitis D antibody; anti-HEV, hepatitis E antibody; HAV, hepatitis A virus; HBV, hepatitis B virus; HDV, hepatitis D virus; HDVAg, hepatitis D antigen; IgG, immunoglobulin G; IgM, immunoglobulin M.

Prevention and Treatment of Viral Hepatitis

Desired Outcomes

- ❖ Prevent the spread of infection
- ❖ Prevent and treat symptoms
- ❖ Suppress viral replication
- ❖ Normalize hepatic aminotransferases
- ❖ Improve liver histology
- ❖ Decrease morbidity and mortality by preventing cirrhosis, hepatocellular carcinoma, and end stage liver disease.

Prevention and Treatment of Viral Hepatitis

General Approach

- Managing viral hepatitis involves both prevention and treatment.
- Prevention of hepatitis A and B (D) with immune globulin or vaccines (no HCV vaccine).
- Acute viral hepatitis is primarily managed with supportive care.
- Mild to moderate symptoms rarely require hospitalization
- Liver transplantation may be required rarely if fulminant hepatitis develops.
- Treatment is available for chronic HBV, HCV, and HDV

Prevention of Viral Hepatitis

Hepatitis A Prevention

- Good personal hygiene and proper disposal of sanitary waste to prevent fecal-oral transmission of HAV.
 - Frequent hand washing with soap & water after using the bathroom and prior to eating meals.
- Drinking bottled water in areas where HAV is most endemic.
- Avoiding fruits, vegetables, and raw shellfish harvested from sewage-contaminated water
- Individuals at high risk of acquiring HAVserum immune globulin (IG) or HAV vaccine.
- Note: Acute HAV treatment is supportive... there is no approved drug therapy.

Hepatitis A Prevention

Immune Globulin (IG)

- contains antibodies from pooled human plasma (usually from >1000 donors)
- provides passive immunization against various infectious diseases, including HAV.
- IG is available in intravenous (IVIG) or intramuscular (IGIM) formulations
 - but only IGIM is used for prevention of HAV.
- IGIM is effective in providing pre- and post-exposure prophylaxis against HAV
 - but it does not confer lifelong immunity.
- IG is most effective if given in the incubation period of the infection
- Adverse effects of IGIM are rare; however, anaphylaxis in individuals who have immunoglobulin A deficiency.hence, IgA deficient patients should not receive IGIM.

Hepatitis A Prevention

Immune Globulin

- Dosing of IG is the same for adults and children.
- For post-exposure prophylaxis and for short-term pre-exposure coverage of < 3 months
 - a single dose of 0.02 mL/kg IM is given.
- For long-term pre-exposure prophylaxis of less than or equal to 5 months
 - a single dose of 0.06 mL/kg is used.
- Either the deltoid or gluteal muscle may be used.
- In children younger than 24 months, IG can be given in the anterolateral thigh muscle

Hepatitis A Prevention

HAV Vaccine

- HAV vaccine (Havrix; Vaqta) is an inactivated virus administered via IM for adults and children >1 year of age.
- **Twinrix** is a preparation for adults that contains immunizations for HAV & HBV in one vaccine.
- HAV vaccine is recommended for
 - Children
 - Persons traveling to countries with endemic HAV
 - Men who have sex with men
 - Chronic liver disease patients
 - Intravenous drug users
 - People with occupational exposure to HAV.
- For pre-travel vaccination... the HAV vaccine should be administered at least 2 weeks prior to expected exposure.

Recommended Dosing of Hepatitis A Vaccines

Vaccine	Age (Years)	Dose of Hepatitis A Antigen (Volume)	No. of Doses	Schedule ^a
HAVRIX	1-18	720 ELISA units (0.5 mL)	2	0, 6-12 months
	≥ 19	1,440 ELISA units (1 mL)	2	0, 6-12 months
VAQTA	1-18	25 units (0.5 mL)	2	0, 6-18 months
	≥ 19	50 units (1 mL)	2	0, 6-18 months
TWINRIX ^b	≥ 18	720 ELISA units (1 mL)	3	0, 1, 6 months
	≥ 18 (accelerated schedule)	720 ELISA units (1 mL)	4	0, 7 days, 21-30 days, +12 months

^aZero (0) denotes initial dose, subsequent numbers denote time after initial dose for timing of additional doses.

^bCombination hepatitis A and B vaccine, also contains 20 mcg of hepatitis B surface antigen and requires a three-dose schedule for adequate HBV response.

ELISA, enzyme-linked immunosorbent assay.

Source: Centers for Disease Control and Prevention.⁶

Hepatitis B Prevention

- Hepatitis B Immune Globulin (HBIG)

- ❖ A sterile solution containing antibodies (high concentration of anti-HBs).
- ❖ A single dose intramuscular of HBIG 0.06 mL/kg provides passive immunization.

- HBV vaccines

- ❖ (Engerix-B; Recombivax HB; Heplisav B)...inactivated virus administered via IM.
- ❖ Twinrix is for adults that contains immunizations for HAV & HBV in one vaccine.
- ❖ Individuals exposed to HBV should receive the hepatitis B vaccine with or without HBIG, preferably within 24 hours of exposure

Prevention.... Hepatitis B

❑ Hepatitis B vaccine is indicated for

- Children
- Adults with occupational exposure to HBV
- Household and sexual contact of HBV carriers
- Men who have sex with men
- People with multiple heterosexual partners

- patients diagnosed with HIV
- Intravenous drug users
- Infants born to mothers with HBV
- Patients on hemodialysis,
- Patients with chronic liver disease

Recommended Dosing of Hepatitis B Vaccines (Single Antigen)

Vaccine	Age (Years)	Dose (Volume)	No. of Doses	Schedule ^a
Engerix-B	0-19	10 mcg IM (0.5 mL)	3	0, 1, 6 months
	>20	20 mcg IM (1 mL)	3	0, 1, 6 months
Recombivax	1-19	10 mcg IM (0.5 mL)	3	0, 1, 6 months
	>20	10 mcg IM (1 mL)	3	0, 1, 6 months
Heplisav-B ^b	≥18	0.5 mL	2	0, 1 months

^aZero (0) denotes initial dose, subsequent numbers denote time after initial dose for timing of additional doses.

^bUtilizes different adjuvant; cannot be interchanged with other HBV vaccines.

Source: Centers for Disease Control and Prevention.⁶

Vaccine Adverse Reactions

- Adverse reactions to the HAV and HBV vaccines include:
 - ❖ Soreness
 - ❖ Warmth, and erythema at the injection site,
 - ❖ Drowsiness
 - ❖ Headache
 - ❖ Fever
 - ❖ Rare cases (<1%) of Guillain-Barre syndrome (GBS) have been reported.
- The only contraindication to HAV and HBV vaccines includes hypersensitivity to vaccines
- HAV and HBV vaccines are pregnancy category C.

Drug Therapy for Chronic HBV

Generally Accepted Criteria for Treatment of HBV

Characteristics:

Anyone with HBeAg (+) or HBeAg (-) active HBV defined as:

- HBV DNA >2,000 IU/mL (kIU/L)
- ALT $2 \times$ upper limit of normal^a and/or with evidence of histological disease^b

Anyone with compensated or decompensated cirrhosis with HBV DNA >2,000 IU/mL (kIU/L)^c

Anyone not fulfilling above criteria with ALT < $2 \times$ ULN and any detectable HBV DNA (<2,000 IU/mL [kIU/L]), consider:

- Patient's age
- Family history of HCC
- Prior history of HBV treatment
- Extrahepatic manifestations of HBV

^aPer European guidelines, any elevations in ALT.

^bModerate or greater fibrosis as determined by biopsy or noninvasive measures.

^cPer European guidelines, any detectable HBV DNA.

Drug Therapy for Chronic HBV

- HBV infections are not curable
- Goals of therapy
 - ❖ To suppress HBV replication and prevent disease progression to cirrhosis and HCC.
 - ❖ Preventing HBV reactivation in patients with inactive HBV infections.
 - ❖ HBe-Ag loss, with or without seroconversion to anti-HBeAg in HBeAg positive
 - ❖ Patients Normalization of ALT is another goal associated with viral suppression.
 - ❖ Loss of HBsAg with or without anti-HBs seroconversion

Drug Therapy for Chronic HBV

	Hepatitis B
Treatment	➤ Pegylated-interferon α-2a (Pegasys) or ➤ Nucleotide Analogs: lamivudine, adefovir, entecavir, telbivudine, tenofovir
Duration	➤ Pegylated-interferon α-2a (Pegasys): 48 wk ➤ Nucleotide Analogs: 6-12 mo after HBeAg seroconversion if HBeAg positive patient or until loss of HBsAg if HBeAg negative patient

Drug Therapy for Chronic HBV

- ❖ Because hepatic damage is sustained by ongoing viral replication, drug therapy aims to suppress viral replication by either immuno-modulating agents or antivirals—the nucleos(t)ide agents (NAs).
- ❖ The immune-mediating agents approved for use are pegylated (peg) interferon (IFN)-alfa injectables.
- ❖ The antiviral agents are all oral therapies and many also have some activity against HIV (entecavir, tenofovir, lamivudine, and telbivudine); thus, it is important to understand a patient's HIV status before starting therapy.
- ❖ All NAs carry a boxed warning for lactic acidosis.

Drug Therapy for Chronic HBV

- The preferred NAs are those considered to have a high barrier to resistance and include [entecavir](#), [tenofovir diprovixil](#), and [tenofovir alafenamide](#).
- Other available NAs which have a low barrier to resistance and are thus not preferred include lamivudine, telbivudine, and adefovir.
- A major difference in therapy between IFN and the NAs is duration of use: IFN-based therapies are typically administered for a predefined duration, whereas NAs are used until a specific end point is achieved.
- The key disadvantage and limitation with IFN-based therapy are the side effect profile and laboratory abnormalities associated with its use.
- In contrast, the NAs are well tolerated and can be safely used even in patients with decompensated cirrhosis.

Hepatitis C : Treatment

❖ No vaccines for prevention

❖ Goal of therapy

- To eradicate HCV infection.
- Virologic cure (non-detectable HCV RNA at least 12 weeks after completing therapy)
- To prevent the development of chronic HCV infection

Hepatitis C : Treatment

- Current standard of care for chronic HCV patients: daily oral tablets for 8 to 12 weeks
- Sustained virologic response rates: 99–100%.
- Sustained virological response (no detectable viral load 12 weeks after conclusion of therapy) is high for all first-line therapies

Hepatitis C : Treatment

- ❖ The treatment of chronic HCV was revolutionized with the approval of Direct Acting Antivirals (DAAs).
- ❖ Previously the treatment backbone included the injection of peg-IFN and was associated with a substantial SE profile.
- ❖ The current standard of care for all chronic HCV infections, regardless of genotypes (GTs), is an all-oral drug regimen.
- ❖ Despite the change in pharmacological therapy, many patients are aware of the substantial SEs associated with IFN-based therapy and are wary of HCV therapy.
- ❖ It is important to clearly distinguish current therapies from IFN-based ones for patients who are familiar with IFN treatment.
- ❖ Recommended therapeutic regimens for treatment-naïve patients and treatment experienced(TE) patients based on genotype, cirrhosis and non-cirrhosis pattern.

AASLD/IDSA Recommended Treatment Regimens for Treatment-Naïve Patients with Hepatitis C Genotypes 1-3

HCV Genotype	No Cirrhosis	Compensated Cirrhosis (CTP Class A)
1a	Elbasvir/Grazoprevir × 12 wks ^a Glecaprevir/Pibrentasvir × 8 wks Ledipasvir/Sofosbuvir × 12 wks ^b Sofosbuvir/Velpatasvir × 12 wks	Elbasvir/Grazoprevir × 12 wks ^a Glecaprevir/Pibrentasvir × 12 wks Ledipasvir/Sofosbuvir × 12 wks Sofosbuvir/Velpatasvir × 12 wks
1b	Elbasvir/Grazoprevir × 12 wks Glecaprevir/Pibrentasvir × 8 wks Ledipasvir/Sofosbuvir × 12 wks ^b Sofosbuvir/Velpatasvir × 12 wks	Elbasvir/Grazoprevir × 12 wks Glecaprevir/Pibrentasvir × 12 wks Ledipasvir/Sofosbuvir × 12 wks Sofosbuvir/Velpatasvir × 12 wks
2	Glecaprevir/Pibrentasvir × 8 wks Sofosbuvir/Velpatasvir × 12 wks	Glecaprevir/Pibrentasvir × 12 wks Sofosbuvir/Velpatasvir × 12 wks
3	Glecaprevir/Pibrentasvir × 8 wks Sofosbuvir/Velpatasvir × 12 wks	Glecaprevir/Pibrentasvir × 12 wks Sofosbuvir/Velpatasvir × 12 wks ^c

CTP, Child-Turcotte Pugh.

^aIf no NS5A resistance detected. ^b8-week duration may be considered for patients who are not black, not HIV coinfectd, and with an HCV RNA <6 million IU/mL (kIU/L). ^cPretreatment resistance testing recommended.

AASLD/IDSA Recommended Treatment Regimens for Treatment-Experienced Patients with Hepatitis C Genotypes 1-3

	<i>Peg-Interferon/Ribavirin TE</i>		<i>NS3 PI + Peg-Interferon TE</i>	
HCV Genotype	Non-Cirrhotic	Compensated Cirrhosis	Non-Cirrhotic	Compensated Cirrhosis
1	Elbasvir/Grazoprevir × 12 wks ^a Glecaprevir/Pibrentasvir × 8 wks Ledipasvir/Sofosbuvir × 12 wks Sofosbuvir/Velpatasvir × 12 wks	Elbasvir/Grazoprevir × 12 wks ^a Glecaprevir/Pibrentasvir × 12 wks Sofosbuvir/Velpatasvir × 12 wks	Ledipasvir/Sofosbuvir × 12 wks Glecaprevir/Pibrentasvir × 12 wks Sofosbuvir/Velpatasvir × 12 wks	Glecaprevir/Pibrentasvir × 12 wks Sofosbuvir/Velpatasvir × 12 wks
2	Glecaprevir/Pibrentasvir × 8 wks Sofosbuvir/Velpatasvir × 12 wks	Sofosbuvir/Velpatasvir × 12 wks Glecaprevir/Pibrentasvir × 12 wks		
3	Sofosbuvir/Velpatasvir × 12 wks ^b	Elbasvir/Grazoprevir plus Sofosbuvir × 12 wks Sofosbuvir/Velpatasvir/Voxilaprevir × 12 wks		

^aFor GT 1a: Recommended treatment if no resistance for NS5A detected.

^bResistance testing for Y93 recommended before use; if Y93 present, add ribavirin or use different regimen.

Hepatitis C : Treatment

- The Direct Acting Antivirals (DAAs) act on one of three targets on the HCV virion.
- The DAAs must be used in combination with at least two drugs acting on different targets.
- The drug nomenclature identifies where the drug is inhibiting viral replication:
 - N3/4A protease inhibitors end with –previr;
 - NS5A inhibitors end with –asvir; and
 - NS5B polymerase inhibitors end with –buvir.
- Early DAAs acted on specific HCV genotypes.
- Sofosbuvir/velpatasvir, glecaprevir/pibrentasvir, and sofosbuvir/velpatasvir/voxilaprevir are pangenotypic, meaning they are effective in all HCV genotypes.
 - In resource limited areas, the ability to treat without pretreatment genotype testing favors the use of these agents and is an increasingly appealing strategy because it can streamline HCV treatment and reduce costs.
- As a group, the DAAs are well tolerated and cause few laboratory abnormalities.

Hepatitis D Prevention and Treatment

- Hepatitis B vaccination can indirectly prevent hepatitis D infections.
- The recommended treatment for HDV is pegIFN(180 mcg SC once wkly) for 48 weeks.
- For mild liver disease, monitor HDV RNA and ALT levels every 12 to 24 weeks.
- Treatment should be considered for individuals with advanced liver disease.
- First-line oral agents for HBV infections (eg, tenofovir) may be co-administered with pegIFN
- Monotherapy with nucleoside/nucleotide analogs is ineffective in reducing HDV DNA levels but may decrease hepatitis B viral loads.

Hepatitis E Prevention and Treatment

- Good personal hygiene and proper disposal of sanitary waste are the most effective ways to prevent viral acquisition.
- There are no commercially available vaccines in the United States to prevent hepatitis E.
- However, a hepatitis E vaccine has been licensed in China since 2011 and may become available for countries with high HEV infection incidence rates.

Evaluation of Therapeutic Outcome

- Monitoring for efficacy in patients treated for chronic hepatitis B or C includes:
 - Evaluating aminotransferase levels
 - Hepatitis B or C viral levels and serologies.
- Monitor for drug adverse effects

End!